

Hello Airflow — Your First DAG

Hello_airflow.py

```
from airflow.decorators import dag, task
from datetime import datetime

@dag(
    dag_id="hello_airflow",
    schedule="@once",
    start_date=datetime(2024, 1, 1),
    catchup=False,
    tags=["Assignment-1", "hello", "Uday"],
)
def hello_airflow():

    @task()
    def say_hello():
        name="Uday"
        print(f"Hello from Airflow! My name is {name}")

    @task()
    def show_date(**context):
        ds = context['ds']
        print(f"Today's execution date is: {ds}")

    say_hello() >> show_date()

hello_airflow()
```

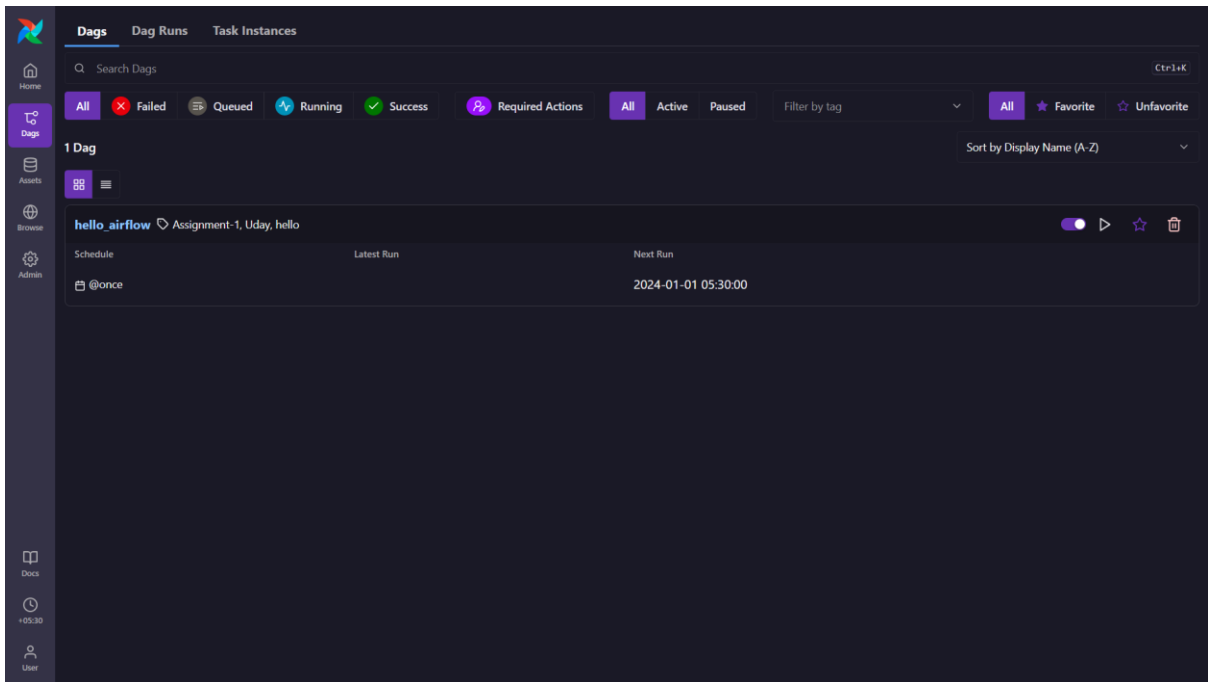
1. Screenshot 1:

- a. Astro CLI terminal — showing astro dev start completed successfully with 3 containers running

```
PS D:\KenexAi\Airflow> cd .\Assignment-1\
PS D:\KenexAi\Airflow\Assignment-1> astro dev start
✓ Project image has been updated
✓ Project started
Warning: could not start proxy: proxy daemon is not supported on Windows
➤ Airflow UI: http://assignment-1.localhost:6563
➤ Postgres Database: postgresql://localhost:5432/postgres
➤ The default Postgres DB credentials are: postgres:postgres
PS D:\KenexAi\Airflow\Assignment-1> █
```

2. Screenshot 2:

Airflow UI DAG list — showing hello_airflow with the toggle switched ON (unpaused)



The screenshot displays the Airflow web interface's DAG list. The left sidebar contains navigation links: Home, DAGs (selected), Assets, Browse, and Admin. The top navigation bar shows 'Dags', 'Dag Runs', and 'Task Instances'. Below this is a search bar and a status filter bar with options: All, Failed, Queued, Running, Success, Required Actions, Active, Paused, and Filter by tag. The 'All' filter is selected. The main content area shows '1 Dag' and a table for 'hello_airflow' with the description 'Assignment-1. Uday, hello'. The table has columns for 'Schedule', 'Latest Run', and 'Next Run'. The 'Schedule' column shows '@once'. The 'Next Run' column shows '2024-01-01 05:30:00'. A toggle switch is visible next to the DAG name, which is currently turned ON (unpaused). Other icons for favorite, unfavorite, and delete are also present.

Schedule	Latest Run	Next Run
@once		2024-01-01 05:30:00

3. Screenshot 3:

Graph View — both task boxes say_hello and show_date showing GREEN (success)

The screenshot shows the Airflow web interface in Graph View. The DAG 'hello_airflow' is displayed with two tasks: 'say_hello' and 'show_date'. Both tasks are green and labeled 'success'. The right-hand pane shows the logs for the 'say_hello' task, indicating it completed successfully at 2026-04-16 18:27:54.

4. Screenshot 4:

Task Log of show_date — showing the execution date printed in the log output

The screenshot shows the Airflow web interface in Task Log View for the 'show_date' task. The task is highlighted in blue and labeled 'success'. The right-hand pane shows the logs for the 'show_date' task, indicating it completed successfully at 2026-04-16 18:27:55. The log output includes the message 'Today's execution date is: 2026-04-16'.

DAG Configuration — Parameters & Scheduling

Dag_config.py

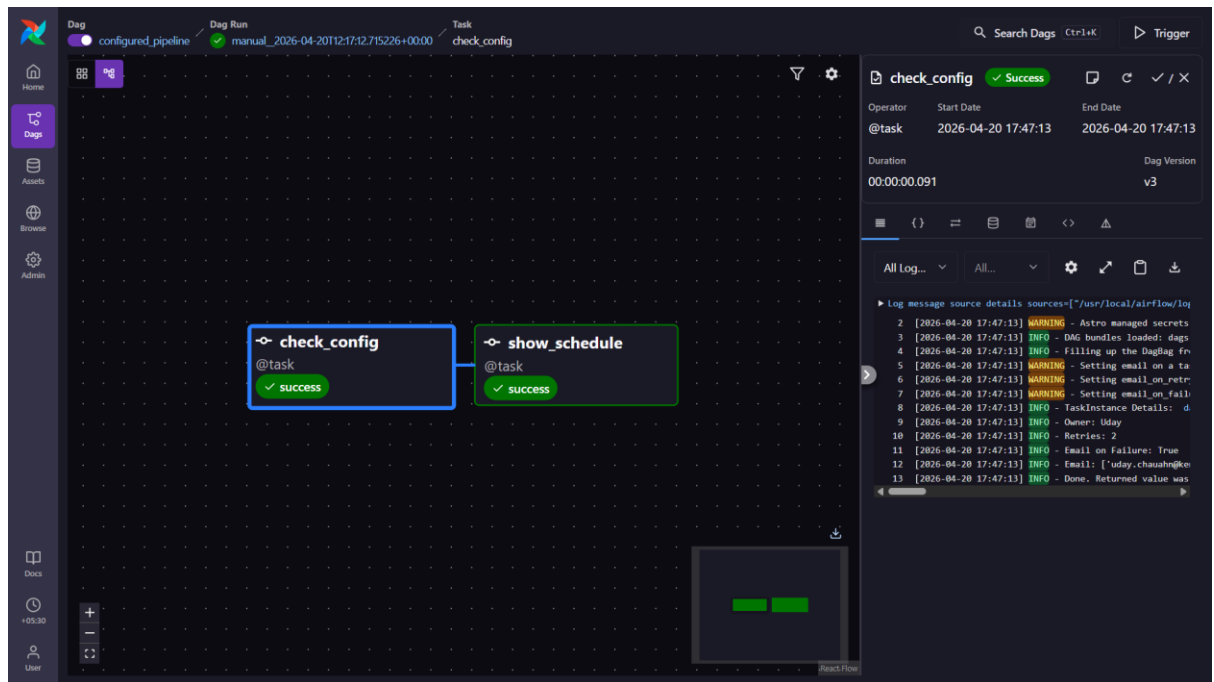
```
from airflow.sdk import dag, task
from datetime import datetime, timedelta

@dag(
    dag_id="configured_pipeline",
    schedule="0 8 * * *", # Run at 8:00 AM every day
    start_date=datetime(2024, 1, 1),
    catchup=False,
    default_args={
        "owner": "Uday",
        "retries": 2,
        "retry_delay": timedelta(minutes=5),
        "email_on_failure": False,
        "email": ["uday.chauahn@kenexai.com"]
    },
    tags=["Assignment-1", "configured", "Uday"],
)
def dag_config():
    @task()
    def check_config(task_instance=None, **context):
        task=task_instance.task
        print(f"Owner: {task.owner}")
        print(f"Retries: {task.retries}")
        print(f"Retry Delay: {task.retry_delay}")
        print(f"Email on Failure: {task.email_on_failure}")
        print(f"Email: {task.email}")

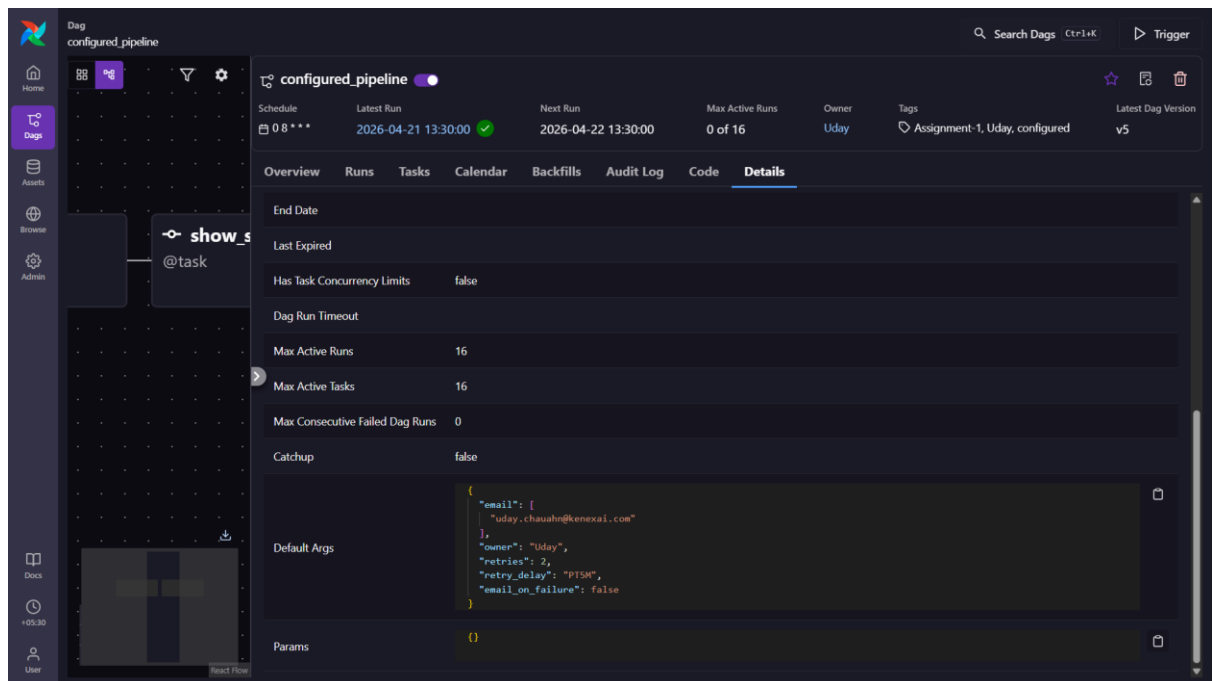
    @task()
    def show_schedule(**context):
        # schedule=context['schedule']
        ds=context['ds']
        print(f"This pipeline runs every day at 8:00 AM")
        print(f"Execution date: {ds}")

    check_config()>>show_schedule()
configured_pipeline = dag_config()
```

1. Graph View — both tasks GREEN showing correct order:
check_config → show_schedule



2. DAG Details panel — showing the schedule "0 8 * * *", start date, and catchup = False



3. Task Log of check_config — showing all 4 default_args values printed correctly

Home

Dags

Assets

Browse

Admin

Docs

+05:30

User

Dag

configured_pipeline

Dag Run

manual_2026-04-21T09:38:19.544339+00:00

Task

check_config

Search Dags

Ctrl+K

Trigger

check_config

Success

Operator

@task

Start Date

2026-04-21 15:08:19

End Date

2026-04-21 15:08:20

Duration

00:00:00.168

Dag Version

v5

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

All Sources

Log message source details sources=["/usr/local/airflow/logs/dag_id-configured_pipeline/run_id-manual_2026-04-21T09:38:19.544339+00:00/task_id-check_2026-04-21 15:08:19.544339+00:00"]

2 [2026-04-21 15:08:19] WARNING - Astro managed secrets backend is disabled

3 [2026-04-21 15:08:19] INFO - DAG bundles loaded: dags-folder

4 [2026-04-21 15:08:19] INFO - Filling up the DagBag from /usr/local/airflow/dags/dag_config.py

5 [2026-04-21 15:08:20] WARNING - Setting email on a task is deprecated; please migrate to SmtplNotifier. category=UserWarning

6 [2026-04-21 15:08:20] WARNING - Setting email on a task is deprecated; please migrate to SmtplNotifier. category=UserWarning

7 [2026-04-21 15:08:20] WARNING - Setting email_on_failure on a task is deprecated; please migrate to SmtplNotifier. category=UserWarning

8 [2026-04-21 15:08:20] INFO - TaskInstance Details: dag_id=configured_pipeline task_id=check_config dagrun_id=manual_2026-04-21T09:38:19.544339

9 [2026-04-21 15:08:20] INFO - Done. Returned value was: None

10 [2026-04-21 15:08:20] INFO - Owner: Uday

11 [2026-04-21 15:08:20] INFO - Retries: 2

12 [2026-04-21 15:08:20] INFO - Retry Delay: 0:05:00

13 [2026-04-21 15:08:20] INFO - Email on Failure: False

14 [2026-04-21 15:08:20] INFO - Email: ['uday.chauhan@kenexai.com']

_config

show_scheduled

@task

success

React Flow

Sequential ETL Pipeline — Tasks & Dependencies

Simple_etl.py

```
from airflow.sdk import dag, task
from datetime import datetime, timedelta

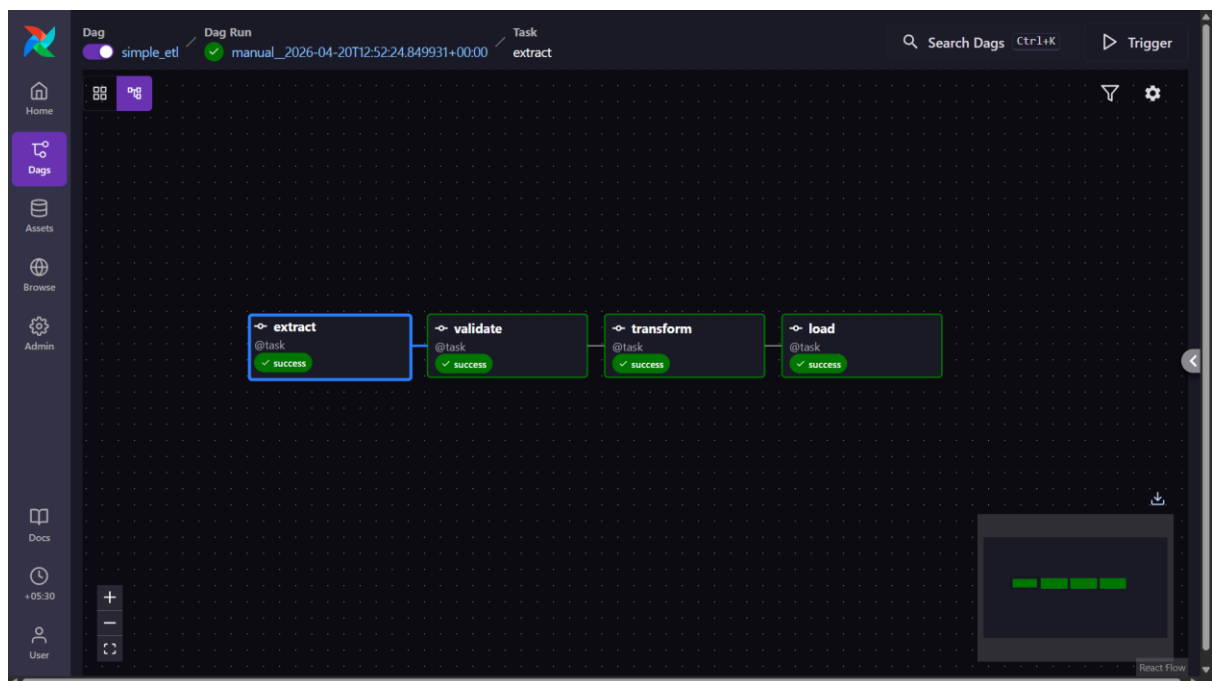
@dag(
    dag_id="simple_etl",
    schedule="@daily",
    catchup=False,
    tags=["Assignment-1", "simple_etl", "Uday"]
)
def simple_etl():
    @task()
    def extract(**context):
        products=[
            {"product_id": 1, "name": "Product A", "price": 10.0},
            {"product_id": 2, "name": "Product B", "price": 20.0},
            {"product_id": 3, "name": "Product C", "price": 30.0},
            {"product_id": 4, "name": "Product D", "price": 40.0},
        ]
        print(f"Extracting sales data for date: {context['ds']}")
        for i in products:
            print(f"Product ID: {i['product_id']}, Name: {i['name']}, Price: {i['price']}")
        return products
    @task()
    def validate(products):
        print("Validating data...")
        print(f"Validation passed – {len(products)} records found")
        return products
    @task()
    def transform(products):
        print("Transforming data...")
        print("Applying 10% discount to all records")
        print("Transformed data:")
        for i in products:
            i['price'] = i['price'] * 0.9
            print(f"Product ID: {i['product_id']}, Name: {i['name']}, Discounted Price: {i['price']}")
        return products
    @task()
    def load(products, **context):
        print("Loading to warehouse...")
```

```

        print(f"Successfully loaded {len(products)} records on
{context['ds']}")
        for i in products:
            print(f"Product ID: {i['product_id']}, Name: {i['name']}, Final
Price: {i['price']}")
            data = extract()
            validated = validate(data)
            transformed = transform(validated)
            load(transformed)
simple_etl = simple_etl()

```

1. Graph View — all 4 tasks GREEN in correct left-to-right order:
extract → validate → transform → load



2. Task Log of transform — showing each product printed with its discounted price

The screenshot shows the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main area displays a DAG named 'simple_etl' with a task named 'transform' highlighted. The task status is 'Success'. The task log is open, showing the following details:

Operator	Start Date	End Date	Duration	Dag Version
@task	2026-04-21 15:24:01	2026-04-21 15:24:01	00:00:00.188	v2

The log messages are as follows:

```
Log message source details sources=["/usr/local/airflow/logs/dag_id=simple_etl/run_id=manual__2026-04-21T09:53:59.147318+00:00/task_id=transform"]
2 [2026-04-21 15:24:01] WARNING - Astro managed secrets backend is disabled
3 [2026-04-21 15:24:01] INFO - DAG bundles loaded: dags-folder
4 [2026-04-21 15:24:01] INFO - Filling up the DagBag from /usr/local/airflow/dags/simple_etl.py
5 [2026-04-21 15:24:01] INFO - TaskInstance Details: dag_id=simple_etl task_id=transform dagrun_id=manual__2026-04-21T09:53:59.147318+00:00
6 [2026-04-21 15:24:01] INFO - Done. Returned value was: [{"name": "Product A", "price": 9.0, "product_id": 1}, {"name": "Product B", "price": 18.0, "product_id": 2}, {"name": "Product C", "price": 27.0, "product_id": 3}, {"name": "Product D", "price": 36.0, "product_id": 4}]
7 [2026-04-21 15:24:01] INFO - Pushing xcom ti=RuntimeTaskInstance(id=8UID('019daf75-ff78-740a-bab2-4e75d8ebefaa'), task_id='transform', dag_id='simple_etl', run_id='manual__2026-04-21T09:53:59.147318+00:00')
8 [2026-04-21 15:24:01] INFO - Transforming data...
9 [2026-04-21 15:24:01] INFO - Applying 10% discount to all records
10 [2026-04-21 15:24:01] INFO - Transformed data:
11 [2026-04-21 15:24:01] INFO - Product ID: 1, Name: Product A, Discounted Price: 9.0
12 [2026-04-21 15:24:01] INFO - Product ID: 2, Name: Product B, Discounted Price: 18.0
13 [2026-04-21 15:24:01] INFO - Product ID: 3, Name: Product C, Discounted Price: 27.0
14 [2026-04-21 15:24:01] INFO - Product ID: 4, Name: Product D, Discounted Price: 36.0
```

3. Task Log of load — showing the "Successfully loaded" message with date and record count

The screenshot shows the Apache Airflow web interface. On the left, a sidebar contains navigation links for Home, Dags, Assets, Browse, and Admin. The main area displays a DAG named 'simple_etl' with a task named 'load' highlighted. The task status is 'Success'. The task log is open, showing the following details:

Operator	Start Date	End Date	Duration	Dag Version
@task	2026-04-21 15:24:02	2026-04-21 15:24:02	00:00:00.157	v2

The log messages are as follows:

```
Log message source details sources=["/usr/local/airflow/logs/dag_id=simple_etl/run_id=manual__2026-04-21T09:53:59.147318+00:00/task_id=load/att"]
2 [2026-04-21 15:24:02] WARNING - Astro managed secrets backend is disabled
3 [2026-04-21 15:24:02] INFO - DAG bundles loaded: dags-folder
4 [2026-04-21 15:24:02] INFO - Filling up the DagBag from /usr/local/airflow/dags/simple_etl.py
5 [2026-04-21 15:24:02] INFO - TaskInstance Details: dag_id=simple_etl task_id=load dagrun_id=manual__2026-04-21T09:53:59.147318+00:00 map_index=0
6 [2026-04-21 15:24:02] INFO - Loading to warehouse...
7 [2026-04-21 15:24:02] INFO - Successfully loaded 4 records on 2026-04-21
8 [2026-04-21 15:24:02] INFO - Product ID: 1, Name: Product A, Final Price: 9.0
9 [2026-04-21 15:24:02] INFO - Product ID: 2, Name: Product B, Final Price: 18.0
10 [2026-04-21 15:24:02] INFO - Product ID: 3, Name: Product C, Final Price: 27.0
11 [2026-04-21 15:24:02] INFO - Product ID: 4, Name: Product D, Final Price: 36.0
12 [2026-04-21 15:24:02] INFO - Done. Returned value was: None
```

Dependency Patterns — Fan-Out & Fan-In

Dependencies.py

```
from airflow.sdk import dag, task
from datetime import datetime

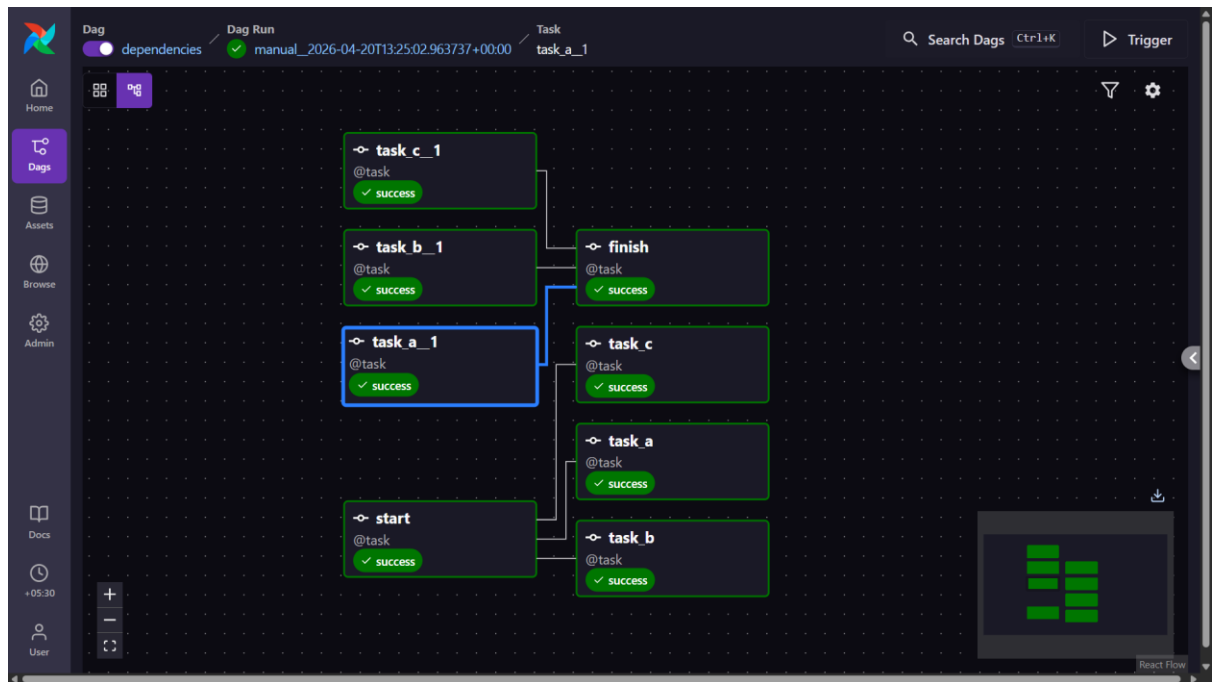
@dag(
    dag_id="dependencies",
    schedule="@daily",
    start_date=datetime(2024, 1, 1),
    catchup=False,
    tags=["Assignment-1", "dependencies"],
)
def dependencies():
    @task()
    def start(**context):
        print("Pipeline started – checking all systems")
        print(f"Execution date: {context['ds']}")

    @task()
    def task_a(**context):
        print("Task A running – processing customer data")
    @task()
    def task_b(**context):
        print("Task B running – processing order data")
    @task()
    def task_c(**context):
        print("Task C running – processing product data")

    @task()
    def finish(**context):
        print("Pipeline finished – all tasks completed")
        print(f"Execution date: {datetime.now()}")
    start() >> [task_a(), task_b(), task_c()]

    [task_a(), task_b(), task_c()] >> finish()
dependencies = dependencies()
```

1. Graph View — showing the diamond shape: start → 3 parallel tasks at same level → finish (all GREEN)



2. Task Log of finish — showing the completion message and datetime.now() timestamp

The screenshot shows the Apache Airflow task log for the 'finish' task. The task is marked as 'Success'. The log messages are as follows:

Log Level	Message	Timestamp
WARNING	Astro managed secrets backend is disabled	2026-04-20 18:55:05
INFO	DAG bundles loaded: dags-folder	2026-04-20 18:55:05
INFO	Filling up the DagBag from /usr/local/airflow/dags/dependencies.py	2026-04-20 18:55:05
INFO	TaskInstance Details: dag_id=dependencies task_id=finish dagrun_id=manual__2026-04-20T13:25:02.963737+00:00 map_index=0	2026-04-20 18:55:05
INFO	Pipeline finished - all tasks completed	2026-04-20 18:55:05
INFO	Done. Returned value was: None	2026-04-20 18:55:05
INFO	Execution date: 2026-04-20 13:25:05.429044	2026-04-20 18:55:05